

Catherine Awad

Software Engineer with 5+ years of programming experience across AI/ML, mobile, full-stack, and embedded systems
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Skills

Languages: Java, Python, Dart, Scala, C, C++, JavaScript, SQL

Frameworks & Tools: Flutter, Spring Boot, Node.js/Express, PyTorch, scikit-learn, Hugging Face Transformers, Firebase, MongoDB, PostgreSQL, Supabase, AWS, Docker, Git, Claude Code, Figma

AI/ML: Deep Learning, Computer Vision, NLP, LLMs, RAG, LoRA, Transfer Learning, model evaluation (pass@k, human/annotator studies)

Hardware/Embedded: Verilog, SystemVerilog, SAMD21/Arduino firmware, LoRa, Assembly (x86, MIPS)

Spoken Languages: Arabic (native), English (fluent)

Education

- **M.S. Computer Science & Engineering**, Santa Clara University (GPA: 3.87) Jan 2025 – Dec 2026 (expected)
- **B.S. Computer & Communications Engineering**, Alexandria University Graduated 2019

Experience

AI Research Assistant, AIM Lab, Santa Clara University May 2026–Present

- Building **MedicalQA**, an LLM system that generates caregiver question–answer pairs from dementia-care videos, by extending the lab’s MentorQA framework; evaluating generation quality across model pipelines.
- Co-developed **McMiner-in-the-Loop**, a misconception-aware tutoring method that feeds automated misconception detection into an LLM tutor’s next turn; evaluated with pass@k on the EvoCodeBench dataset by difficulty and round on a fault-tolerant GPU harness.

Software Engineering Research Assistant, EPIC Lab, Santa Clara University 2025–Present

- Developing **Hydration Automation**, an IoT platform for real-time monitoring of agricultural hydration systems, across its Flutter app, Node.js/Express backend, and Arduino/C++ firmware; set up the CI pipeline, commit hooks, design-system checks, and integration docs the team builds against.
- Rebuilding **IncidentAid**, a firefighter incident-reporting and response-coordination app, from its earlier native versions into a single cross-platform Flutter app — new UI/UX (prototyped in Figma), Riverpod state management, and role-based dashboards.

Co-Founder & Lead Engineer, Frona Tech, Egypt July 2021 – Dec 2023

- Led a team of 4 engineers building **Frona**, a cross-platform (Android/iOS) diet and fitness app for meal/workout planning and calorie, macro, and workout tracking.
- Took the product from concept to release on the App Store and Google Play; built with Scala, gRPC, Flutter, AWS, and Docker.

Teaching Assistant, Computer Science Department, Alexandria University Sept 2019 – Jan 2020

- Taught classes and labs in Programming 1, Design of Algorithms, and Operating Systems.

Selected Projects

MedicalQA — Caregiver QA Generation & Evaluation (LLMs, RAG, React, Supabase)

- Creating MedicalQA by extending the AIM Lab’s MentorQA framework: contributed the Teepa Snow dementia-care dataset and reproducibility controls (seeding, temperature, decoding config) across its single-agent, multi-agent-chunking, topic-segmentation, and RAG pipelines.
- Built a human-annotation web app (React/Vite + Supabase, Netlify) and analyzed human judgments from the supported training videos to validate the evaluation rubric.

McMiner-in-the-Loop — Misconception-Aware LLM Code Tutoring (Python, pass@k evaluation)

- Bridged two systems that each fell short alone: TRAVER tutors through dialogue but cannot detect misconceptions, while McMiner detects misconceptions from code but cannot tutor. Every dialogue round generates diagnostic code, mines the student’s specific false belief, and injects it into the tutor’s prompt so the next hint targets that misconception.
- Evaluated with pass@1/5/10 on the EvoCodeBench dataset by difficulty and round; the analysis also surfaced an over-tutoring effect, where added rounds help weaker students but degrade stronger ones.
- Built a resilient evaluation harness (auto-pause/retry across multi-hour model-server outages) on GPU pods, plus a demo web app showing baseline vs. McMiner-in-the-loop dialogues side by side.

Medical Anomaly Detection using Vision-Language Models (CLIP, LoRA)

- Few-shot medical anomaly detection with CLIP and LoRA fine-tuning on RSNA (26k+ chest X-ray images).
- Implemented multiple anomaly-scoring methods; improved performance via model fusion and regularization.

Hydration Automation — IoT Agricultural Monitoring (Flutter, Node.js, Arduino/C++)

- Built the Flutter app (NFC- and QR-based device onboarding, real-time sensor dashboards, localization with dark mode) and the SAMD21/LoRa firmware: ACK-gated transmission with offline queueing, NTP-disciplined clocks, OTA-refreshed device whitelists, and battery telemetry; co-developed the backend.

IncidentAid — Firefighter Incident Reporting & Coordination (Flutter, Firebase, Figma)

- Riverpod state management, a repository-pattern data layer, and a Realtime Database → Firestore migration behind role-based dashboards; integrated Google Maps/Places and BLE beacon scanning, with prod/dev flavors on isolated Firebase environments for App Store release.

Volunteer Service Management App (Flutter, Firebase, Riverpod)

- Built solo, from scratch, for the volunteer service in Egypt listed under Leadership: a bilingual (Arabic/English) app that organizes records of the visually impaired and elderly community members served, plus volunteers, groups, and announcements.
- Role-based access with admin/moderator approval flows, audit logging, timezone-aware push notifications (FCM + Cloud Functions), and compressed/cached image handling on Firebase (Firestore, Auth, Storage, Crashlytics); live on Google Play.

BeyondList — Collaborative Shopping & Expense App (Flutter + Spring Boot / React Native)

- Real-time collaborative shopping-list and expense management for roommates and family.
- Built in two stacks to broaden full-stack skills: an original Flutter + Firebase (Firestore, Bloc/Cubit, push notifications) version, then a rewrite with a Java Spring Boot backend (Spring Security OAuth2/JWT, JPA, PostgreSQL) and a React Native (Expo, Zustand) frontend.

Garden Simulation System (Java, JavaFX)

- Modular simulation of plant growth and environmental conditions, with clean separation between simulation logic, UI, and API (garden management, watering, pest control, temperature regulation).

Bachelor's Graduation Thesis — Multi-Speaker TTS (Tacotron 2)

- Implemented Google's Tacotron 2 for multi-speaker text-to-speech and speech synthesis.
- Added an Arabic (Egyptian dialect) dataset and fine-tuned the model to generate realistic speech.

Leadership & Volunteer

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- 15+ years volunteering with elderly people, orphans, and people with disabilities in Egypt. 2009–Present
 - Led training and coordination of volunteers supporting the visually impaired and elderly at the same organization in Egypt. 2020–Present
 - Class Representative, Alexandria University — represented peers, facilitated faculty communication, resolved issues, coordinated activities. 2016–2019